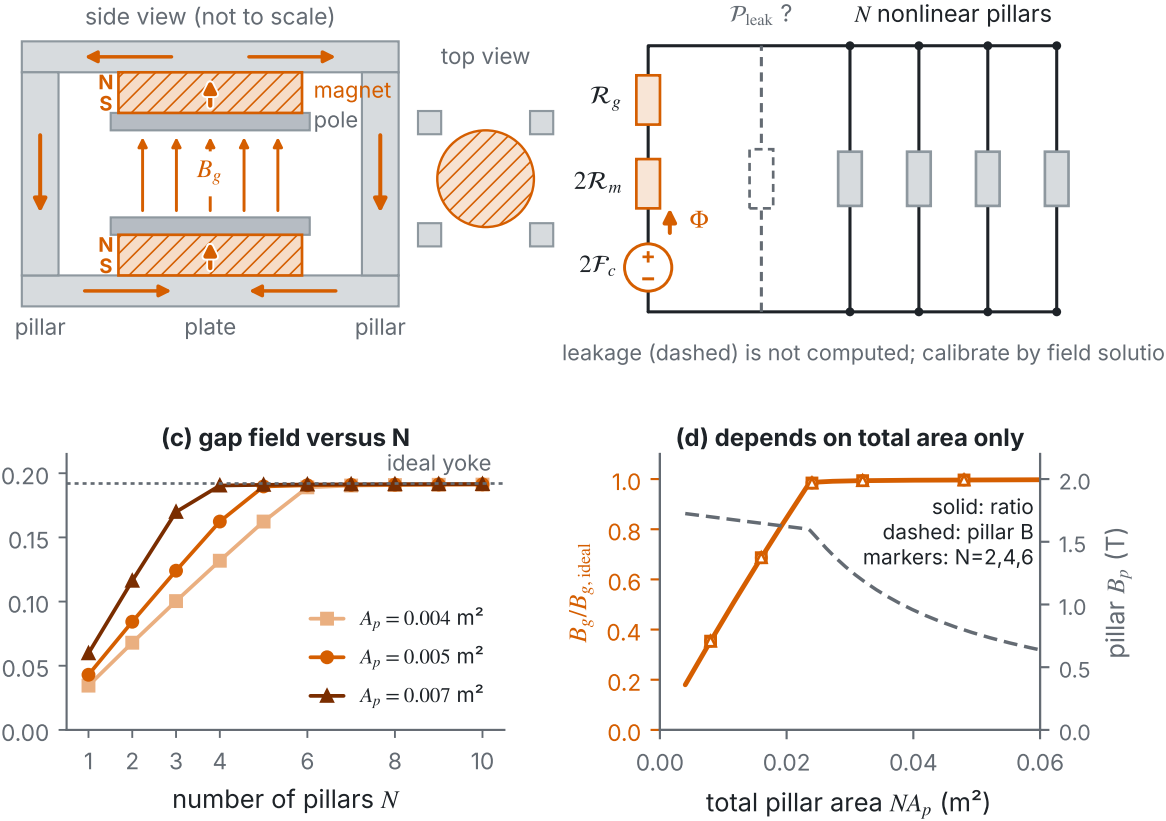
side view (not to scale)
N
S
magnet
pole
B_g
N
S
pillar
plate
pillar
top view
$\mathcal{P}_{\text{leak}}$?
N nonlinear pillars
$\mathcal{R}_g$
$2\mathcal{R}_m$
Φ
$2\mathcal{F}_c$
leakage (dashed) is not computed; calibrate by field solution
(c) gap field versus N
ideal yoke
B_g (T)
0.20
0.15
0.10
0.05
0.00
$A_p = 0.004$ m²
$A_p = 0.005$ m²
$A_p = 0.007$ m²
1
2
3
4
6
8
10
number of pillars N
(d) depends on total area only
solid: ratio
dashed: pillar B
markers: N=2,4,6
$B_g/B_{g,\text{ideal}}$
1.0
0.8
0.6
0.4
0.2
0.0
pillar B_p (T)
2.0
1.5
1.0
0.5
0.0
0.00
0.02
0.04
0.06
total pillar area NA_p (m²)